Cognitive Resume Fit Analyzer (CRFA)

- An AI-Based Resume Screening System
  - Capstone Project - Artificial Intelligence
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Problem Statement

- Manual resume screening is time-consuming and biased.
  - Recruiters struggle with large applicant volumes.
  - Need for intelligent and automated evaluation.

Project Objective

- Develop an AI-driven resume evaluation framework.
  - Rank candidates based on multi-dimensional scoring.
  - Provide explainable and data-driven insights.

System Architecture

- Input Layer: Job Description + Resumes
  - Processing Layer: NLP + Transformer + Scoring Engine
  - Output Layer: Ranking + AI Fit Score Visualization

Technology Stack

- Python
  - Google Colab
  - Sentence-Transformers
  - Pandas, NumPy, Matplotlib

Data Processing Pipeline

- Text Cleaning and Normalization
  - Experience Extraction using Regex
  - Skill Keyword Identification

Semantic Intelligence Layer

- Transformer Model: all-MiniLM-L6-v2
  - Generates contextual embeddings
  - Cosine similarity for semantic matching

Multi-Dimensional Scoring Model

- 1. Semantic Similarity Score (50%)
  - 2. Skill Relevance Score (30%)
  - 3. Experience Score (20%)
  - Hybrid weighted AI evaluation framework

Skill Scoring Mechanism

- Weighted Skill Matrix
  - Verilog, FPGA, Digital Design
  - ASIC, Python, Machine Learning

Experience Quantification

- Years of experience extracted via regex.
  - Normalized scoring scale (0–1).
  - Adds real-world evaluation depth.

Final AI Fit Score Formula

- Final Score = (0.5 × Semantic)
  - + (0.3 × Skill)
  - + (0.2 × Experience)
  - Ranks candidates intelligently.

Results & Ranking

- Candidates ranked based on AI Fit Score.
  - DataFrame output generated.
  - Visualization using bar graph.

Key Features

- ✓ Transformer-Based NLP
  - ✓ Explainable AI scoring
  - ✓ Modular Architecture
  - ✓ Online Deployable (Colab)

Project Implementation Link

- Google Colab Notebook:
  - https://colab.research.google.com/drive/17huT2C845vfXxMD2A3Nl8qVsxGaEcwWY?usp=sharing

Conclusion & Future Scope

- Efficient automated hiring assistance.
  - Can be extended to real resume PDFs.
  - Future: Deploy as web application using Streamlit.